ESHS SERIES
MODULAR FUHRER

LiFePO4 (lithium iron phosphate) battery compatible

ONLINE DOUBLE CONVERSION UPS

10KVA ~ 520KVA (1/1 - 3/1 - 3/3)

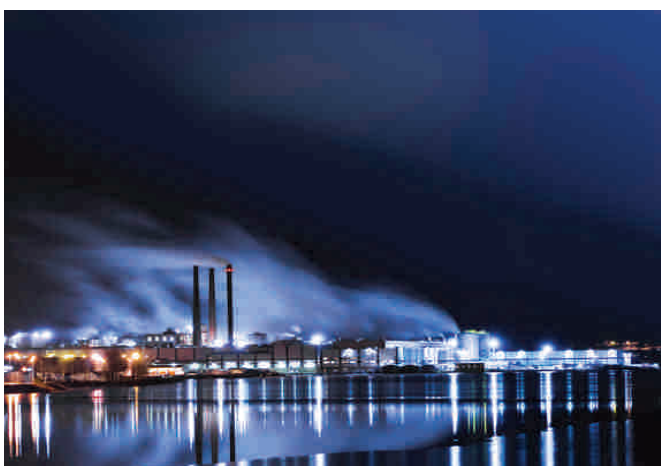
Modular Redundant Design

Ultimate Protection
Mission Critical &
I.T Grade UPS

ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

DP Electronics (Deutsche Power Co., Limited)







ESHS Series (1/1 - 3/1 - 3/3) 10KVA ~ 520KVA

UPS For Mission Critical & IT, Industrial Equipments

ESHS Series Fuhrer is a Multi-Standard Modular High frequency technology Parallel redundant UPS system. Fuhrer UPS set of today's most advanced technology application achievement, has the most advanced and sophisticated technology, with innovative excellence concept focusing on the development of the global power, in order to push UPS power fields of technological progress and the birth of high-end products.

This innovative state of the art scalable, redundant UPS system focuses on elevating the users' experience to new heights. With the highest efficiency rate of 96% and a significantly smaller footprint, the product offers a leading solution in the range of up to 500 KW/KVA.

Modular design

All units adopt modular design, including power module, byp ass module, monitoring module, can be easily int egrated in MDC or customized cabinet Power module, Bypass module, Monitoring module, ECU control module, all these modules ar e hot-swappable.



ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

Energy Saving

Compared with traditional UPS, which efficiency less than 90% about 10% is wasted. Fuhrer inverter efficiency is about 96%. It is designed for lower heat emission, less energy waste.

Application

Large mainframe oriented data centers Computer rooms Small mainframe Mini computers Centralized or clustered servers Telecommunications applications Medical analysis equipment such as MRI and CAT scanners Laboratory instrumention Mission critical customized applications.

Hot Swappable

Hot swappable module can be added or removed online. There is no need for technical personal to perform the job . It is safe and convenient for maintenance and repair as no need to bypass the UPS.

Parallel System

Fuhrer modular UPS has the characteristics that it has the independent power and control sections. Each module is an dependent UPS. It adopts the centralized multiple parallel N+X format. It's simple for system expansion with little fault points. It provides load with N+1 to N+X levels of protection. The maximum can be expanded to 100kva.

Easy Expansion

With the upgrade of customers' load requirement the original designed UPS may not be able to meet the demand. According to the traditional UPS system, we must replace with a new and higher capacity UPS system for customers. But for Fuhrer modular UPS there is no need to replace the whole UPS, but only adding the required number of modules for expansion, it saves customer's costs for initial equipment purchasing and future expansion.

LBS function

LBS function can realize 2 independent UPS system work in synchr onization, and it enhances the reliability of the systems.

Intelligent recharging system

Fuhrer UPS system applies a two step intelligent charging system. The first stage is a constant charging current that can recharge the battery capacity to 90% very quickly. The system then transfers to a constant voltage mode to guarantee the battery can stay fully charged all the time.
This intelligent charging system not only reduces the battery recharging time but also extends battery life, saving on battery replacement costs.

High capacity MTBF

System MTBF for two modules in parallel is more than 1,000,000 hours and power availability is above 99.999%. Each redundant configuration guarantees correct operation even in the event of the failure of one of the UPS modules. The module replacement procedure only takes 5 minutes for full system recovery. This solution allows you to: • Minimize downtime; • Reduce the number of stored spare parts; • Avoid the need for specialized technicians.

Standard

Safety: IEC / EN62040-1,
IEC / EN60950-1
EMC: IEC / EN62040-2,
IEC61000-4-2, IEC61000-4-3,
IEC61000-4-4, IEC61000-4-5,
IEC61000-4-6, IEC61000-4-8
Noise: EN62040-2





Multi-setting

According to input power condition, user can set up the UPS configuration with 4 kinds of working model (3 phase in & 3phase out, 3 phase in & 1 phase out, single phase in & out, single phase in & 3 phase out).

Bypass module

High reliability, overload capacity: 100% for one minute The switching time of less than 0ms Possess very strong post, interlock, protection, and other functions Equipped with USB, RS232, RS458, communication function Replacement does not affect the power supply to the load.

VRLA & Lithium battery supportable

VRLA batteriy number of each group can be selected from 30pcs to 50pcs (continuously adjustable).

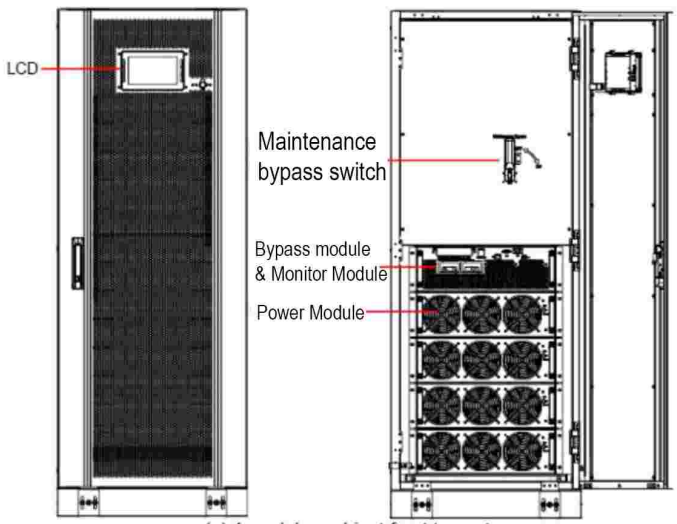
Match with Deutsche Power lithium battery rack, providing higher power density, lower footprint and longer cycle life Configuration of VRLA or Lithium can be chose from LCD Two wire connection, simplify the construction on site and save the cost of battery neutral cable.

Intelligent management

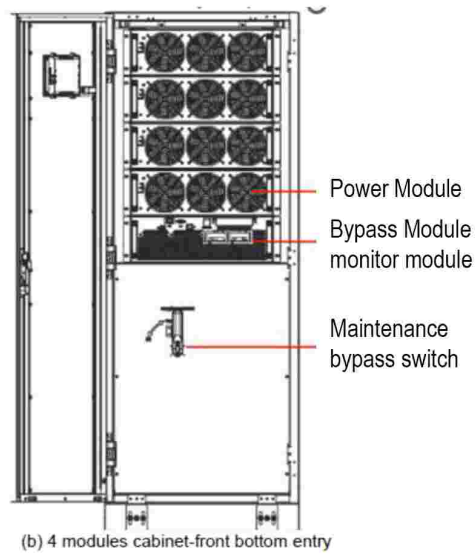
·Standard colorful touch screen
·Support recording and exporting history logs and fault logs
·Support SNMP, RS232, RS485, BMS, Dr y contact interface
·Support upgrade of CAN of power module inside of c abinet

DP Electronics (Deutsche Power Co., Limited)

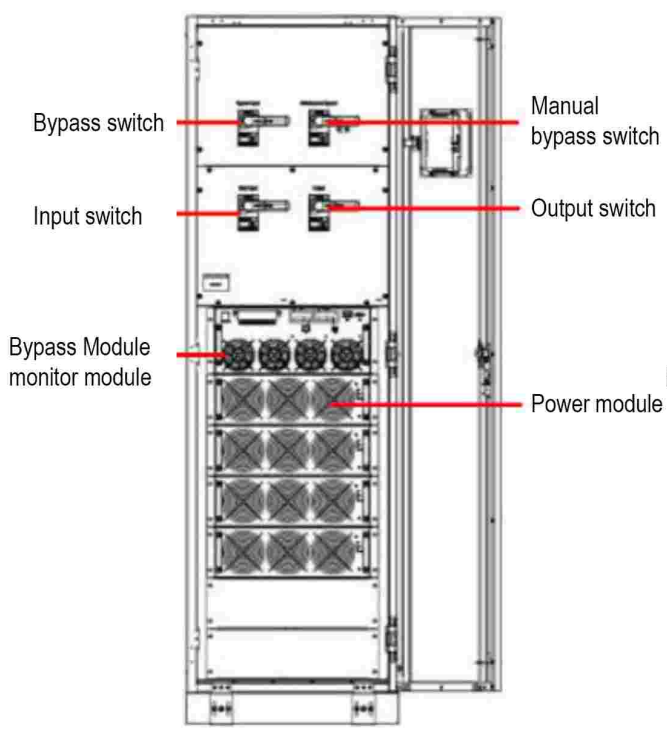
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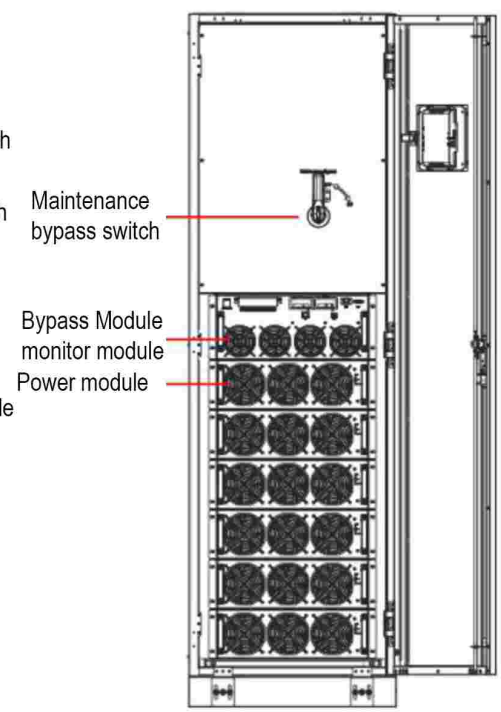
(a) 4 modules cabinet-front top entry



(b) 4 modules cabinet-front bottom entry



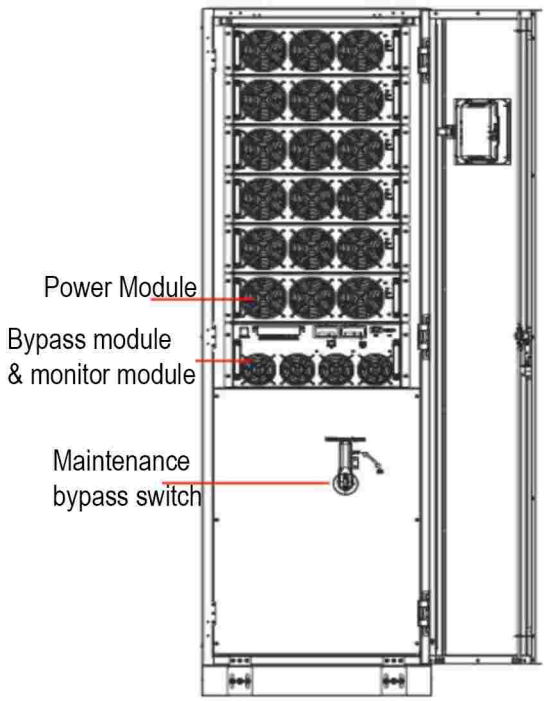
(c) 4 modules cabinet with 4 switches



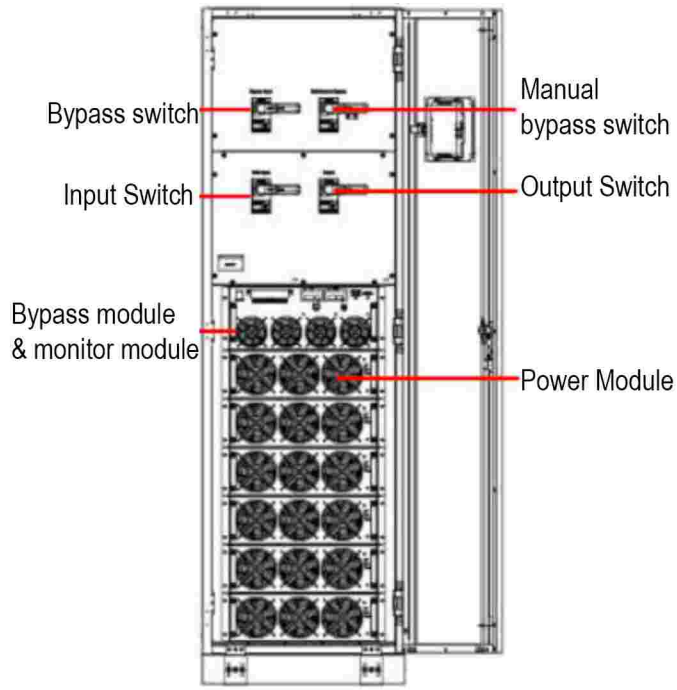
(d) 6 modules cabinet-front top entry

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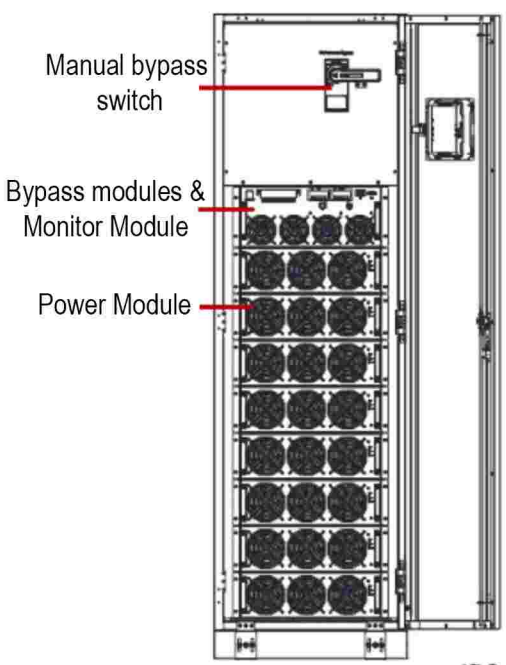
ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

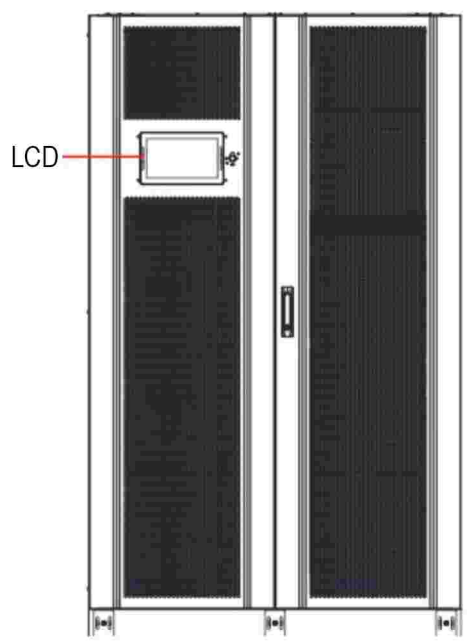


(e) 6 modules cabinet-front bottom entry



(e) 6 modules cabinet with 4 switches

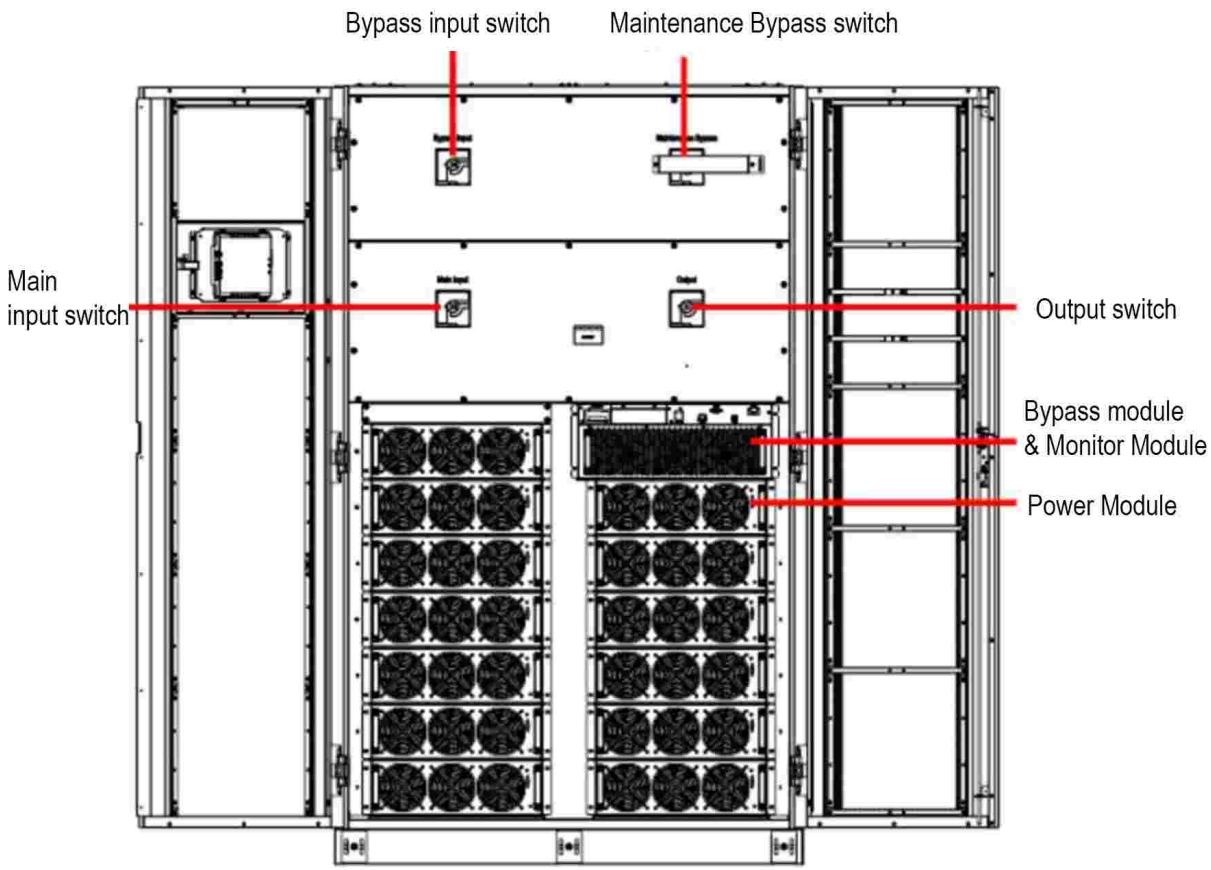




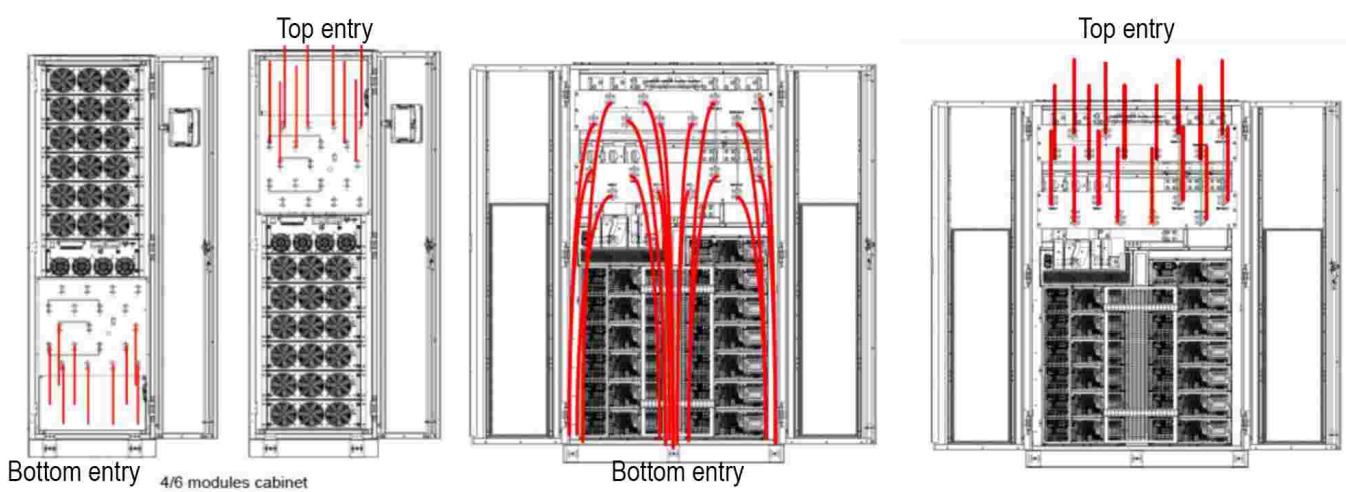
(f)8 modules cabinet

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(q)10 & 12 modules cabinet



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ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

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TECHNICAL SPECIFICATION

Module Model		ES-1033-HS-RT		
Cabinet Model		FS3300E-20	FS3300E-40	FS3300E-60
Cabinet capacity (VA)		10k~20k	10k~40k	10k~60k
Module capacity (VA)		10k		
Max. Number		2	4	6
Module Model		ES-1533-HS-RT		
Cabinet Model		FS3300E-30	FS3300E-60	FS3300E-90
Cabinet capacity (VA)		15k~30k	15k~60k	15k~90k
Module capacity (VA)		15k		
Max. Number		2	4	6
Module Model		ES-2033-HS-RT		
Cabinet Model		FS3300E-40	FS3300E-80	FS3300E-120
Cabinet capacity (VA)		20k~40k	20k~80k	20k~120k
Module capacity (VA)		20k		
Max. Number		2	4	6
Module Model		ES-2533-HS-RT		
Cabinet Model		FS3300E-50	FS3300E-100	FS3300E-150
Cabinet capacity (VA)		25k~50k	25k~100k	25k~150k
Module capacity (VA)		25k		
Max. Number		2	4	6
Module Model		ES-3033-HS-RT		
Cabinet Model		FS3300E-60	FS3300E-120	FS3300E-150
Cabinet capacity (VA)		30k~60k	30k~120k	30k~150k
Module capacity (VA)		30k		
Max. Number		2	4	5+1
INPUT				
Nominal voltage		380/400/415Vac, (3Ph+N+PE)		
Operating voltage range		138~305Vac for 40% Load; 305~485Vac for 100% Load;		
Operating frequency range		40Hz-70Hz		
Power factor		≥0.99		
Harmonic distortion (THDi)		≤3% (100% linear load)		
Bypass voltage range		Max. Voltage: 220V: +25% (optional+10%, +15%, +20%); 230V: +20% (optional +10%, +15%); 240V: +15% (optional +10%)		
		Min. voltage: -45% (optional-10%, -15%, -20%, -30%)		
Bypass frequency range		Frequency protection range: ±10%		
Power Walk In		Support		
Generator input		Support		
OUTPUT				
Rated voltage		380/400/415Vac, (3Ph+N+PE)		
Power factor		1		
Voltage regulation		±1%		
Output frequency	Line Mode	Synchronize with input, when the input frequency > ±10% (±1%/±2%/±3%/±4%/±5% optional), output 50/60 (±0.1Hz)		
	Bat. Mode	(50/60±0.1%)Hz		
Crest factor		3:01		

ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

TECHNICAL SPECIFICATION

Harmonic distortion (THDv)		≤ 1% with linear load; ≤3% with nonlinear load		
Efficiency		up to 95.8%		
BATTERY				
Battery voltage		Optional Voltage: ±180/192/204/216/228/240/252/264/276/288/300Vdc(30/32/34/36/38/40/42/44/46/48/50pes optional);		
		360Vdc-600Vdc (30-50 pes, 36 pes default, 36-50 pes no power derating; 32-34 pes output power factor 0.9; 30 pes output power factor 0.8)		
Power module charge current		18A (Max.)		
SYSTEM FEATURES				
Transfer time		Utility to Battery : Oms; Utility to bypass: Oms		
Overload	Line Mode	≤110%, 60min; ≤125%, 10min; ≤150%,1min; to bypass.> 150% Shut down Immediately.		
	Bypass Mode	135% overload for long term; >1000% overload for 100 ms		
Overheat		Line Mode: Switch to Bypass; Backup Mode: Shut down UPS immediately		
Low battery voltage		Alarm and Switch off		
Self-diagnostics		Upon Power On and Software Control		
Back-feed		Support		
EPO		Shut down UPS immediately (turn to bypass optional)		
Battery		Advanced Battery Management		
Noise suppression		Complies with EN62040-3		
Audible & Visual alarms		Line Failure, Battery Low, Overload, System Fault		
Status LED & LCD display		Line Mode, Bypass Mode, Battery Low, Battery Fault, Overload & UPS Fault		
Reading on the LCD display		Input, Output, Battery, Command, Setting, Maintenance		
Communication inter face		RS232, RS485, Parallel, LBS, BMS, Dry contact port, Relay card(optional), SNMP card(optional), Battery temperature sensor(optional)		
ENVIRONMENTAL				
Operating temperature		0℃ – 40℃		
Storage temperature		-25°C- 55℃		
Humidity range		0 – 95% (noncondensing)		
Altitude		<1500m, derating required when >1500m		
Noise level(from 1M distance)		<58dB	<60dB	<62dB
PHYSICAL				
Dimension WxDxH (mm)	UPS cabinet	485x850x353(8U)	485x850x575(13U)	485x850x752(17U)
	Power module	440x620x86(2U)		
Net weight (kg)	UPS cabinet	69	79	98
	Power module	10kVA: 19; 15–30kVA: 21		
STANDARDS				
Safety		IEC/EN62040-1, IEC/EN62477-1		
EMC		IEC/EN62040-2 (IEC61000-4-2, IEC61000-4-3, IEC61000-4-4, IEC61000-4-5, IEC61000-4-6, IEC61000-4-8)		
Specifications are subject to change without prior notice.				

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ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

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TECHNICAL SPECIFICATION

MODEL		ES-XX-33-HSRT-80/120/200/320	ES-XX-33-HSRT-200/300/400/500/600/800/1000	ES-XX-33-HSRT-300/600/840/1080/1200
Capacity (VA/W)	UPS Cabinet	80~320k	200~1000k	300~1200k
	Module	40	50	60
	Max. Number	2/3/5/8	4/6/8/10/12/16/20	5/10/14/18/20
Max. Parallel Number		8		
INPUT				
Nominal Voltage (Vac)		380/400/415, (3Ph+N+PE)		
Operating Voltage Range (Vac)		138~305 for 40% load; 305~485 for 100% load		
Power Factor		≥0.99		
Harmonic Distortion (THDi)		≤3% (100% Linear load)		
Bypass Voltage Range (Vac)		Max. voltage: 220: +25% (Optional+10%, +15%, +20%) 230: +20% (Optional+10%, +15%) 240: +15% (Optional+10%) Min. voltage: -45% (Optional-10%, -15% -20%, -30%)		
Bypass Frequency Range (Hz)		50/60±10%		
OUTPUT				
Nominal Voltage (Vac)		380/400/415 (3Ph+N+PE)		
Voltage Regulation		±1%		
Output Frequency (Hz)		Line mode: Synchronize with input, when the input frequency >±10% (±1%/±2%/±4%/±5% optional), output 50/60 (±0.1z); Bat. mode: (50/60±0.1%)		
Crest Factor		3:1		
Harmonic Distortion (THDv)		≤2% with linear load; ≤4% with nonlinear load		
Overload	Inverter mode	≤110% 60min, ≤125% 10min, ≤150% 1min, >150% 1.2s shut down inverter		
	Bypass mode	30℃: 135% for long term; 40℃: 125% for long term; >1000%, 100ms		
EFFICIENCY*				
AC Mode		Up to 96%		
ECO Mode		Up to 99%		
HECO Mode		Up to 99%		
BATTERY				
Battery Type		VRLA/Li-ion		
Battery Voltage (Vdc)		360~600		
Charging Current (Max.)(A)		20		
MANAGEMENT				
LCD Display		Line voltage, Bypass Mode, Battery Low, Battery Fault, Overload & UPS Fault		
Alarm		Line Failure, Battery Low, Overload, System Fault		
Communication Ports		RS232, RS485, Parallel, LBS, BMS, Dry contact port, Relay card (Optional), SNMP card (Optional), Battery temperature sensor (Optional)		
SYSTEM FEATURES				
Overheat		Line Mode: Switch to Bypass; Backup Mode: Shut down UPS immediately		
Self-diagnostics		Upon Power On and Software Control		
Generator Input		Support		
EPO		Shut down UPS immediately		
ENVIRONMENTAL				
Operating Temperature (℃)		0~40		
Storage Temperature (℃)		-25~55		
Humidity Range		0~95% (Non condensing)		
Altitude (m)		<1000, derating required when >1000		
Noise Level (dB)		<66	<73	
PHYSICAL				
Dimension WxDxH (mm)	UPS Cabinet (S/F)	80/120kVA: 600×850×1200 200/320kVA: 600×850×2000	200/300/400kVA: 600×850×2000 500/600kVA: 1200×850×2000 800/1000kVA: 2000×850×2000	300kVA: 600×850×2000; 600kVA: 1200×850×2000 840/1080kVA: 2000×850×2000; 1200kVA: 2200×850×2000
	Power Module	440×620×130		
Weight (kg)	UPS Cabinet	80kVA: 155; 120kVA: 165; 200kVA: 270; 320kVA: 290	200kVA: 270; 300kVA: 290; 400kVA: 310; 500kVA: 650; 600kVA: 720; 800kVA: 980; 1000kVA: 1080	300kVA: 290; 600kVA: 720; 800kVA: 980; 1080kVA: 1080; 1200kVA: 1200
	Power Module	33	34	35
STANDARDS				
Safety		IEC/EN 62040-1, IEC/EN 62477-1		
EMC		IEC/EN 62040-2 (IEC 61000-2-2, IEC 61000-4-2, IEC 61000-4-3, IEC 61000-4-4, IEC 61000-4-5, IEC 61000-4-6, IEC 61000-4-8, IEC 61000-4-11)		
Performance		IEC 62040-3: 2021, EN IEC 62040-3: 2021		

ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

TECHNICAL SPECIFICATION

MODEL		ESHS-FS100400	ESHS-FS100500	ESHS-FS100600	ESHS-FS100800	ESHS-FS1001000	ESHS-FS1001200
Capacity (VA/W)	UPS Cabinet	400k	500k	600k	800k	1000k	1200k
	Module	100k					
	Max. Number	4	5	6	8	10	12
Max. Parallel Number		6			4		
INPUT							
Nominal Voltage (Vac)		380/400/415 (3Ph+N+PE)					
Operating Voltage Range (Vac)		138~324 for 40% Load; 323~485 for 100% Load;					
Power Factor		≥0.99					
Harmonic Distortion (THDi)		≤3% (100% Linear load)					
Bypass Voltage Range (Vac)		Max. voltage: 220: +25% (Optional+10%, +15%, +20%) 230: +20% (Optional+10%, +15%) 240: +15% (Optional+10%) Min. voltage: -45% (Optional -10%, -15% -20%, -30%)					
Bypass Frequency Range (Hz)		50/60±10%					
OUTPUT							
Nominal Voltage (Vac)		380/400/415, (3Ph+N+PE)					
Voltage Regulation		±1%					
Output Frequency (Hz)		Line mode: ±1%/±2%/±4%/±5%/±10% of the rated frequency (Optional); Bat. mode: 50/60±0.1%					
Crest Factor		3:1					
Harmonic Distortion (THDv)		≤1% with linear load; ≤3% with nonlinear load					
Overload	Inverter Mode	105%~110% overload for 60 min; 110%~125% overload for 10 min; 125%~150% overload for 1 min					
	Bypass Mode	125% overload for long term; >1000% overload for 100 ms					
EFFICIENCY*							
AC Mode		Up to 97%					
ECO Mode		Up to 99%					
HECO Mode		Up to 99%					
BATTERY							
Battery Type		VRLA/Li-ion					
Battery Voltage (Vdc)		360~600					
Charging Current (Max.)(A)		100					
MANAGEMENT							
LCD Display		Input,Output, Battery, Command, Setting, Maintenace					
Alarm		Line Failure, Battery Low, Overload, System Fault					
Communication Ports		RS232, RS485x2 (User commnunication), BAT_T (NTC&RS485), Parallel, LBS, BMS, Dry contact port, Relay card (Optional), SNMP card (Optional), Battery temperature sensor (Optional)					
Dry Contact Ports		"Input (12Vdc): EPO (NO/NC), Battery breaker detection port x3, external maintenance breaker detection port, external output breaker detection port, external bypass breaker detection port, battery grounding detection port, SPD detection port, generator connection detection port and optional port x4. Output: Backfeed (Relay NO/NC), battery breaker tripx3 (24Vdc), optional port (relay)x6"					
SYSTEM FEATURES							
Overheat		Line Mode: Switch to Bypass; Backup Mode: Shut down UPS immediately					
Self-diagnostics		Upon Power On and Software Control					
Generator Input		Support					
EPO		Shut down UPS immediately					
ENVIRONMENTAL							
Operating Temperature (℃)		0~40					
Storage Temperature (℃)		-25~55					
Humidity Range		0~95% (Non condensing)					
Altitude (m)		<1000, derating required when >1000					
Noise Level (dB)		<73		<74		<75	
PHYSICAL							
Dimension WxDxH (mm)	UPS Cabinet(S)	800×1000×2000			800×1000×2000	1400×1000×2000	
	UPS Cabinet(F)				1400×1000×2000	1800×1000×2000	
	Power Module	440×755×130 (3U)					
Weight (kg)	UPS Cabinet(S)	305	330	350	405	690	760
	UPS Cabinet(F)	350	380	410	780	850	920
	Power Module	52.5					
STANDARDS							
Safety		IEC/EN 62040-1, IEC/EN 62477-1					
EMC		IEC/EN 62040-2 (IEC 61000-2-2, IEC 61000-4-2, IEC 61000-4-3, IEC 61000-4-4, IEC 61000-4-5, IEC 61000-4-6, IEC 61000-4-8, IEC 61000-4-11)					
Performance		IEC 62040-3: 2021, EN IEC 62040-3: 2021					

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ESHS SERIES FUHRER ON-LINE DOUBLE CONVERSION UPS

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High capacity MTBF

DEUTSCHE POWER LIFEPO4 High Voltage Key specs include a nominal voltage of 3.2V per cell, energy densities of 40-160 Wh/kg, and cycle lives exceeding 2,000-5,000 charges. They operate efficiently between -20℃ to 60℃ and lack toxic heavy metals, making them safer than other lithium-ion variants.

DEUTSCHE POWER LIFEPO4 series battery system is High voltage system for communications backup type LiFePO4 (lithium iron phosphate) battery products, the system uses the advanced LiFePO4 battery technology with the benefit of long cycle life, small size, light weight, safety and environmental protection, and has a strong environmental adaptability, it is idea for harsh outdoor environments.

The system also integrates a smart battery management and monitoring module, support for remote centralized monitoring and remote battery management and maintenance, to meet the requirements of unattended. Therefore, the DEUTSCHE POWER LIFEPO4 system can fully meet the backup power supply requirements of the access network equipment, mobile communications equipment, transmission equipment, micro base stations and microwave communication equipment.





TECHNICAL SPECIFICATION

Model	CLFP480-40-H	CLFP480-80-H	CLFP480-120-H	CLFP480-40*N-H
Nominal Voltage	480V			
Cluster	1	2	3	N
Adjustable Range Voltage	384/480/499.2/528V			
Normal Capacity	40AH~50AH	40AH~100AH	120AH~150AH	40AH*N
Energy	19.2KWH	38.4KWH	57.6KWH	19.2KWH*N
Cycle Life	5000 times @25C 1C/1C 80% DOD			
Charge Voltage	532V			
Cell Voltage	3.2V			
Adjustable Charge Voltage	426V	532V	553V	585V
Standard Charging Current	40A	80A	120A	40A*N
Max Charge Current	80A	160A	240A	80A*N
Adjustable Discharge Cutt-Off Voltage	375V			
Discharge Current	80A	160A	240A	80A*N
Max Discharge Charge Current	160A	320A	480A	160A*N
Discharge Temperature	20~55C			
Charge Temperature	0~55C			
Storage Temperature	10~45C			
Communication	RS232/RS485/CAN			
Display	LCD			
Terminal	Anderson/Connector			
Weight	280KG	560KG	840KG	280KG*N
Combination mode	21U*1P	21U*2P	21U*3P	21U*NP
Dimensions	IP54			
Certification	CE,UL,ROHS,UN38.3			

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